

Table 3 – Reduced model, logistic regression results of the survival probability of 135 corals, monitored for five consecutive years

Effect	Level	β	SE β	Wald ²	p
Time	Linear	–0.210	0.096	4.79	0.029
Species	<i>Acropora cervicornis</i>	–0.700	0.390	3.22	0.073
	<i>A. palmata</i>	–0.810	0.292	7.70	0.006
	<i>Porites porites</i>	–0.730	0.226	10.44	0.001
Species $\times$ time	<i>A. cervicornis</i> $\times$ time	0.636	0.181	12.32	<0.001
Species $\times$ transplant	<i>A. palmata</i> $\times$ transplant	0.840	0.291	8.32	0.004
Model fit	–2 $\times$ Log likelihood			d.f.	p
Reduced model [*]	420.71			393	<0.001

Logit parameter estimates (β) and standard errors, Wald's chi-square statistics (Wald χ^2) and significance test results (p), and overall reduced model fit are shown. The reduced model excludes those factors that did not significantly contribute to the values of the dependent variable (i.e., survival).

* Significance test results for the overall reduced model fit are relative to a constant time-effects model (i.e., intercept only, base-line hazard does not vary across time periods; Deviance = 451.47, d.f. = 398).

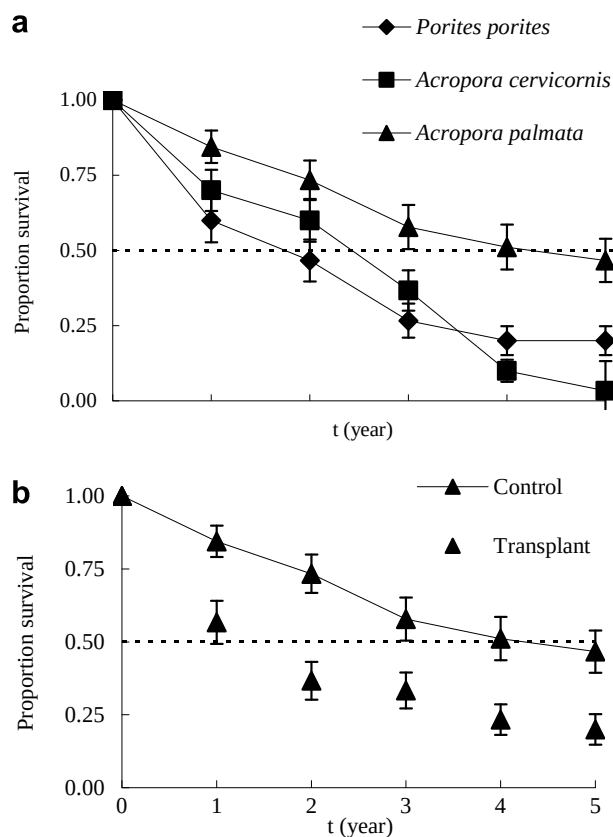


Fig. 4 – (a) Proportion of *Acropora palmata* control colonies, and all *A. cervicornis* and *P. porites* colonies that survived with time, and (b) proportion of *A. palmata* control colonies and transplants that survived across time periods. Standard errors presented with proportion measurements are Greenwood's approximations.

ple size combined with high mortality or the different growth strategies of these species. Some *A. palmata* colonies cemented to the substrate as early as 6-months after transplanting, yet others never cemented after 7 years. Only 20% of *A. cervicornis* and 13% of *P. porites* transplants cemented to the substrate during the study.

3.3. Agents of mortality

Physical dislodgement was the major cause of colony mortality over the 5-year study, accounting for: 58% of both transplanted fragment and control-colony mortality overall; 83% of *P. porites* loss; and, 50% of *A. palmata* and *A. cervicornis* mortality (Fig. 3).

Dislodgement played an even greater role during the first 7 months of the study, causing 93% of both transplant and control-colony mortality, with only 7% attributed to a Category 5 hurricane. Greater survival of transplant than control *P. porites* and *A. cervicornis* colonies in the first 7 months was most likely a result of attachment (Fig. 3); all fragments were attached, whereas most control colonies were not.

Breakage of coral skeleton was commonly observed, usually in association with strong ocean swells. Incidence of breakage was greater in control (17%) than transplant (5%) colonies, presumably due to the larger size of most controls. The extent of breakage ranged from the loss of small branches to removal of a colony's entire vertical structure, leaving only the encrusting base. Some *A. palmata* colonies survived breakage multiple times, resulting in highly transformed colony structures. In a few colonies, serial breakage over 5 years produced a thicket of *A. palmata* clones. Direct damage to monitored colonies by snorkelers or divers was never observed during the study.

Disease-like lesions were observed on *A. palmata* and less often on *A. cervicornis*. Confirmation of disease requires culture or molecular techniques and was outside the scope of this study. However, the causative pathogen of acroporid seriatiosis (*Serratia marcescens*; Patterson et al., 2002) was identified (culture-based techniques) from a single *A. palmata* colony on a study reef (Weil, 2004; Smith, personal communication). Predators feeding on coral tissue were rarely observed on monitored colonies, but predation was inferred based on the pattern of tissue loss and the presence of the predators at the base of a colony. *Coralliophila* spp. were observed feeding on both *Acropora* species. Tissue loss on tips of *A. cervicornis* colonies appeared similar to that produced by *Hermodice carunculata* grazing, but no feeding was directly observed. Because observations in the final 3 years were at 12-month